Arjuna JEE 2025

Mathematics

Straight Lines

DPP: 3

Q1 Find the equation to the locus of the point P if the sum of the square of its distances from $(1,2)$ and $(3,4)$ is 25 units.

Q2 Let A and A' be the points $(5,0)$ and $(-5,0)$ respectively. Find the equation of the locus of all points $\mathrm{P}(x,y)$ such that $|\mathrm{AP}| - |\mathrm{A'P}| = 8$.

Q3 The ends of the hypotenuse of a right angled triangle are $(0,6)$ and $(6,0)$. Find the locus of the third vertex (containing right angle).

Q4 The line passing through the point of intersection of $x+y=2, x-y=0$ and is parallel to $x+2y=5$, is

(A) $x+2y=1$ (B) $x+2y=2$
(C) $x+2y=4$ (D) $x+2y=3$

Q5 The equation of a line passing through the point $(-3,2)$ and parallel to x-axis is

(A) $x-3=0$ (B) $x+3=0$
(C) $y-2=0$ (D) $y+2=0$

Q6 If the slope of a line is 2 and it cuts an intercept -4 on y-axis, then its equation will be

(A) $y-2x=4$ (B) $x=2y-4$
(C) $y=2x-4$ (D) None of these

Q7 The equation of the line inclined at an angle of 60° with x-axis and cutting y-axis at the point $(0,-2)$ is

(A) $\sqrt{3}y = x - 2\sqrt{3}$
(B) $y=\sqrt{3}x-2$
(C) $\sqrt{3}y = x + 2\sqrt{3}$
(D) $y=\sqrt{3}x+2$

Q8 The equation of the straight line which passes through the point $(1,-2)$ and cuts off equal intercepts from axes will be

(A) $x+y=1$ (B) $x-y=1$
(C) $x+y+1=0$ (D) $x-y-2=0$

Q9 If a line passes through the point $P(1,2)$ makes an angle of 45° with the x-axis and meets the line $x+2y-7=0$ in Q, then PQ equals-

(A) $\frac{2\sqrt{2}}{3}$
(B) $\frac{3\sqrt{2}}{2}$
(C) $\sqrt{3}$
(D) $\sqrt{2}$

Q10 The equation of a line parallel to $x+2y=1$ and passing through the point of intersection of the lines $x-y=4$ and $3x+y=7$ is

(A) $x+2y=5$
(B) $4x+8y-1=0$
(C) $4x+8y+1=0$
(D) None of these

Q11 The lines PQ whose equation is $x-y=2$ cuts the x axis at P and Q passes through is $(4,2)$. The line PQ is rotated about P through 45° in the anticlockwise direction. The equation of the line PQ in the new position is

(A) $y=-\sqrt{2}$
(B) $y=2$
(C) $x=2$
(D) $x=-2$

Q12 The point $P(a,b)$ lies on the line $3x+2y=13$ and the point $Q(b,a)$ lies on the line

$4x-y=5$. The equation of the line PQ is
(A) $x-y=5$ (B) $x+y=5$
(C) $x+y=-5$ (D) $x-y=-5$

Q13 If the x intercept of the line $y=mx+2$ is greater than $\frac{1}{2}$ then the gradient of the line lies in the interval
(A) $(-1,0)$
(B) $\left(-\frac{1}{4},0\right)$
(C) $(-\infty,-4)$
(D) $(-4,0)$

Q14 Let $P\equiv(-1,0), Q\equiv(0,0)$, and $R\equiv(3,3\sqrt{3})$ be three points. Then the equation of the bisector of $\angle \mathrm{PQR}$ is
(A) $(\sqrt{3}/2)x+y=0$
(B) $x+\sqrt{3}y=0$
(C) $\sqrt{3}x+y=0$
(D) $x+(\sqrt{3}/2)y=0$

Q15 The equation of the diagonal through origin of the quadrilateral formed by the lines $\mathrm{x}=0, \mathrm{y}=0, \mathrm{x}+\mathrm{y}-1=0$ and $6\mathrm{x}+\mathrm{y}-3=0$ is
(A) $4x-3y=0$
(B) $3x-2y=0$
(C) $x=y$
(D) $x+y=0$

Q16 If the straight line drawn through the point P $(\sqrt{3},2)$ and inclined at an angle $\frac{\pi}{6}$ with the x-axis meets the line $\sqrt{3}x-4y+8=0$ at Q, then the length of PQ is
(A) 3 (B) 4
(C) 5 (D) 6

Q17 The straight lines $7x-2y+10=0$ and $7x+2y-10=0$ form an isosceles triangle with the line $y=2$. The area of this triangle is equal to
(A) $15/7$ sq. units
(B) $10/7$ sq. units
(C) $18/7$ sq. units
(D) None of these

Q18 The perimeter of a parallelogram whose sides are represented by the lines
$x+2y+3=0, 3x+4y-5=0, 2x+4y$
$+5=0$
and $3x+4y-10=0$ is equal to
(A) $\frac{5}{2}+5\sqrt{5}$ units
(B) $5+5\sqrt{5}$ units
(C) $5+\frac{5}{2}\sqrt{5}$ units
(D) $\frac{5+5\sqrt{5}}{2}$ units

Answer Key

Q1 $2x^2 + 2y^2 - 8x - 12y + 5 = 0$

Q2 $9x^2 - 16y^2 = 144$

Q3 $x^2 + y^2 - 6x - 6y = 0$

Q4 (D)

Q5 (C)

Q6 (C)

Q7 (B)

Q8 (C)

Q9 (A)

Q10 (B)

Q11 (C)

Q12 (B)

Q13 (D)

Q14 (C)

Q15 (B)

Q16 (D)

Q17 (C)

Q18 (A)

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